

Poisoning the Threshold-Calibration Stage in Federated IoT Anomaly Detection: A Policy-Differentiated Vulnerability Analysis

Salaheddine Nasloubi¹, Mohamed Lachgar², and Hamid Hrimech¹

¹ LAMSAD Laboratory, National School of Applied Sciences (ENSA), Hassan First University, Berrechid, Morocco

² L2IS Laboratory, Higher Normal School (ENS), University Cadi Ayyad, Marrakech, Morocco

Abstract. Federated learning (FL) defenses for IoT anomaly detection focus on the training phase—Byzantine-robust aggregation, gradient filtering, differential privacy. The post-training *threshold-calibration stage*, where each client autonomously collects a benign calibration set and derives its detection threshold, is structurally outside the scope of these defenses. We characterize the vulnerability of this stage under a score-level contamination model: a single compromised client replaces a fraction f of its calibration buffer with tail-sampled benign reconstruction scores, shifting the detection threshold without altering model weights, training data, or test labels. Across three threshold-policy families (GLOBAL, LOCAL, CLUSTER) evaluated on the N-BaIoT nine-device federation, the score-level raising attack achieves a 100% pass rate on our pre-specified materiality gate (Gate-1; §4) at all non-zero fractions, degrading victim true-positive rates by 2.4 to 7.9 percentage points across injection fractions (4.9–7.9 pp at $f=0.4$). The score-level lowering attack achieves elevated false-alarm rates for LOCAL and CLUSTER, increasing the cross-client false-positive coefficient of variation $\Delta CV(\text{FPR})$ by 0.05–0.06 (a lower bound under REPLACE-FIXED-BUDGET); the $\Delta \text{TPR} > 0$ reflects threshold displacement, not a security benefit, as it increases the alarm burden that may lead operators to disable the detector. The lowering attack is structurally bounded for GLOBAL due to $1/N$ dilution. Policy choice determines not just per-victim magnitude but blast radius: LOCAL confines harm to the compromised client; CLUSTER extends it to co-cluster members; GLOBAL dilutes per-victim shift but propagates it to all federation members. All results are at the score-substitution level: we characterize a necessary calibration-channel vulnerability mechanism, not a complete traffic-generation attack, and end-to-end traffic-level realization is future work.

Keywords: Federated Learning · Anomaly Detection · IoT Security · Poisoning Attack · Threshold Calibration · Vulnerability Analysis

1 Introduction

Federated learning (FL) has emerged as a natural fit for IoT anomaly detection: devices contribute to a shared model without exposing raw traffic, and personalized detection thresholds let each node adapt to its own traffic distribution [13,16]. The dominant security concern in federated anomaly detection focuses on the *training phase*—Byzantine clients injecting adversarial gradients, label-flipping poisoning, or model-inversion attacks [5,4,19]. Aggregation-phase defenses such as FLTrust, Krum, and Trimmed-Mean target exactly this surface [7,6,20].

A post-training coverage gap. What existing defenses collectively ignore is the *threshold-calibration stage*: the post-training, pre-deployment step in which each client collects a window of benign traffic, computes local reconstruction scores, and derives its detection threshold at a fixed quantile q . Training-phase and aggregation-phase defenses are structurally blind to this stage—they complete before calibration begins. If a single client’s benign calibration buffer can be contaminated with semantically benign but anomalously high (or low) reconstruction scores, the resulting threshold shifts without any change to model weights, training data, labels, or gradients. Training- and aggregation-phase defenses do not monitor this post-training calibration stage as evaluated here; they complete before calibration begins and cannot observe it. We make no claim that a dedicated calibration-stage defense is impossible—only that the training- and aggregation-phase defenses studied in the FL-IDS literature do not cover this stage.

This paper. We identify threshold calibration as an under-analyzed attack surface and characterize its vulnerability under a score-level contamination model. Concretely, an adversary with access to a single client’s calibration buffer replaces a fraction f of entries with tail-sampled scores drawn from the same client’s benign distribution (*calibration-stage poisoning*). We evaluate three threshold-policy families—GLOBAL (federation-wide mean), LOCAL (per-client), and CLUSTER (cluster-averaged)—across all nine physical-device clients in the N-BaIoT dataset [15] at four injection fractions.

To preempt the natural objection—“this is just data poisoning applied to calibration data”—we note that calibration-stage poisoning is structurally distinct along three axes (Table 1): it is *temporally isolated* (strictly after model convergence, bypassing training-phase defenses); it exploits a *phase-specific trust model* (calibration data is collected autonomously with no central validation gate); and its effects are *policy-dependent* (the same injection fraction yields qualitatively different threshold shifts and spillover depending on whether thresholds aggregate globally, locally, or by cluster).

Contributions.

1. We isolate the post-training threshold-calibration stage as an attack surface not covered by training-phase and aggregation-phase defenses, distinct from training-data and aggregation-phase poisoning (§3).
2. We present a controlled vulnerability characterization under score-level calibration contamination across three threshold policies, showing that the

score-level raising attack reliably degrades victim detection rates by 2.4 to 7.9 percentage points across non-zero injection fractions (4.9–7.9 pp at $f=0.4$) with 95% bootstrap confidence intervals excluding zero (§5).

3. We document a policy-differentiated vulnerability profile: LOCAL exhibits the largest per-victim threshold shifts with harm confined to the compromised client (no cross-client spillover); CLUSTER exhibits near-LOCAL shifts with potential intra-cluster spillover to co-cluster members; GLOBAL exhibits magnitude-diluted but fleet-wide propagation to all federation members; the score-level lowering attack succeeds for LOCAL and CLUSTER but is conditionally bounded for GLOBAL (§5).

Scope. All claims are bounded to the nine-device N-BaIoT setting with autoencoder detectors and FedAvg aggregation, and the poisoning mechanism operates at the score level; end-to-end traffic-generation realizability is future work (§6). We do not claim deployability, evasion guarantees, or generalization beyond this setting. Defense design against calibration-stage poisoning is out of scope, though we discuss properties a defense would need to satisfy (§6).

2 Background and Related Work

Federated anomaly detection for IoT. Network-level anomaly detection using federated learning allows devices to collaborate on a shared model while keeping raw traffic local [16,17]. Autoencoder-based reconstruction-error detectors are a standard approach: each client trains a local encoder-decoder on benign traffic; anomalies are flagged when reconstruction error exceeds a per-client detection threshold [21]. The N-BaIoT benchmark [15] provides labeled benign and Mirai/BASHLITE traffic for nine physical IoT device types, establishing a widely used evaluation platform for FL-IDS research.

Defenses. Existing defenses target the *training phase*. Byzantine-robust rules, including Krum [6], Trimmed-Mean, and Median [20], filter or downweight outlier gradients before aggregation. FLTrust [7] anchors aggregation to a trusted server dataset. Differential privacy approaches add noise to local updates [10]. These defenses all operate on gradients or model updates; they complete before the threshold-calibration stage begins.

Threshold calibration and poisoning. By construction, a detection threshold at fixed quantile q is sensitive to the scores admitted to the calibration distribution: tail-high contamination raises the boundary (missed detections); tail-low contamination lowers it (false alarms). Poisoning attacks on anomaly detectors have been studied in centralized settings [12,8,3], but these assume either a single data owner or a trusted aggregator who can sanitize training data across the full dataset. Kloft & Laskov [12] study online poisoning of centralized anomaly detectors via insertion into a continuous training stream; their adversarial model requires persistent model-update access, which is absent in the post-training calibration stage of FL. In FL, calibration data never leaves the client, so server-side sanitization is not available. Prior work on FL model poisoning [2] and FL inference attacks [9] focuses on the gradient/weight surface.

Table 1. Positioning relative to adjacent poisoning literature. DATP-CP differs by acting strictly after training, on the autonomously collected benign calibration buffer, leaving model weights, gradients, training data, and test data unchanged.

Prior work type	Attack surface	What changes	Distinction from DATP-CP
Training-data poisoning [5,19]	Training set / labels	Model parameters	DATP-CP leaves training data and model unchanged
FL model poisoning [4,2]	Client gradients	Model weights	Needs no gradient access; acts after convergence
Byzantine aggregation [6,20]	Aggregation inputs	Server aggregate	Does not touch aggregation; harm is post-aggregation
Centralized AD poisoning [12,8]	Central train stream	Decision boundary	No trusted sanitizer exists in FL calibration
Threshold personalization [11,18]	Threshold policy	Per-client/cluster τ	Studies utility, not adversarial contamination
DATP-CP (ours)	Calibration scores	Threshold τ only	Score-level, post-training, single-client

To the best of our knowledge, no prior work characterizes the vulnerability of the post-training threshold-calibration stage in federated threshold-based anomaly detection (Table 1). We do not claim a new class of poisoning in the abstract sense; we empirically show that, in federated threshold-based IoT anomaly detection, post-training calibration contamination creates policy-specific harm profiles that the surveyed training- and aggregation-phase defenses do not monitor.

Threshold policies. Federated threshold personalization spans a spectrum. GLOBAL policies share a single federation-mean threshold; LOCAL policies preserve per-client thresholds; CLUSTER policies group clients by calibration-score similarity and apply cluster-level thresholds [11,18]. The security implications of this design choice have not been studied under calibration-stage adversaries.

3 Threat Model

Attack surface. The threshold-calibration stage occurs after global model convergence and before threshold-dependent detection begins. Each client i collects a benign calibration set $\mathcal{C}_i = \{s_1, \dots, s_{n_i}\}$ of reconstruction scores from its own traffic, then sets its local threshold $\tau_i = \text{quantile}(\mathcal{C}_i, q)$ at a fixed quantile q (here $q=0.95$). Depending on the federation’s threshold policy, τ_i may be used directly (LOCAL), averaged across clients (GLOBAL), or averaged within a calibration-score cluster (CLUSTER).

No existing FL training-phase or aggregation-phase defense monitors or validates the calibration buffer: it is collected autonomously by the client after the aggregation round completes.

Adversary model. The adversary controls a single compromised client v , which we call the *poisoned client*. Under LOCAL policy, only client v 's own detection degrades; under GLOBAL policy, all federation members receive the contaminated threshold. We use “victim” to refer to client v itself when measuring per-client detection harm, and “non-victim clients” when referring to uninvolved federation members.

The adversary has *gray-box score-level access*: since the adversary controls the client, it can read and modify the local calibration buffer before threshold computation. The adversary does not access model weights, modify gradients, alter other clients' data, or touch test data, test labels, or training labels. This is a strictly weaker capability than model-poisoning adversaries, which require gradient injection. The adversary operates at the score level; generating traffic that achieves a target reconstruction score is out of scope (end-to-end realizability is discussed in §6). The REPLACE-FIXED-BUDGET mechanism samples with replacement from the tail reservoir, which at high fractions produces duplicate scores detectable by a uniqueness check (disclosed in §6).

Attack mechanics. We evaluate two attack objectives:

Raise. THRESHOLD-RAISE (primary): for $f > 0$, replace $m = \max(1, \text{round}(fn_v))$ positions in $\mathcal{C}_v$ with scores drawn with replacement from the top-10% tail of $\mathcal{C}_v$ (HIGH-SCORE-BENIGN source). All replacement scores are semantically benign; AUROC is invariant by construction (test scores and labels are unchanged under poisoning). A raised threshold τ_v increases the detection threshold, suppressing alarms at the victim site.

Lower. THRESHOLD-LOWER (secondary, conditional): replace m positions with scores from the bottom-10% tail (LOW-SCORE-BENIGN). A lowered threshold increases the false-positive rate (alarm burden) and FPR disparity; the resulting $\Delta\text{TPR} > 0$ reflects threshold displacement and is a secondary indicator of operating-point shift. This attack is structurally conditional: random position replacement displaces upper-tail support with probability $\approx f$.

Negative control. RANDOM-BENIGN: replacement scores are sampled uniformly from the full calibration distribution rather than a tail. Any effect attributable to score contamination rather than position perturbation should vanish in this condition.

Scope boundary. This paper characterizes the vulnerability at the score level. A complete end-to-end attack would additionally require generating network traffic whose autoencoder reconstruction score lands in the target tail—this is future work (§6). The score-level characterization is a controlled experiment, not a deployed attack: it isolates a necessary calibration-channel vulnerability mechanism, not a complete traffic-generation attack. Claims are bounded to the single-client, single-compromise scenario; multi-client collusion is not evaluated.

Table 2. Experimental configuration for reproducibility.

Component	Configuration
<i>Autoencoder architecture</i>	
Input features	115 (N-BaIoT statistical traffic features)
Encoder layers	115 $\rightarrow$ 80 $\rightarrow$ 40 $\rightarrow$ 20; decoder mirrors
Activation	ReLU (no batch normalization)
Loss	Mean squared reconstruction error
Optimizer	Adam, lr = 0.001, weight_decay = 0
<i>Federated training (FedAvg)</i>	
Local epochs	1
Aggregation	Sample-count-weighted mean
Rounds	40 initial, 150 max (conv.: $\Delta\text{loss} \leq 0.005$ over 10)
Batch size	256
<i>Calibration (per-device sizes in released split manifest)</i>	
Benign samples / device	2,622 (Ecobee) – 35,048 (Philips)
Threshold quantile q	0.95

4 Experimental Protocol

4.1 Dataset and Federation

We use N-BaIoT [15], a labeled IoT network traffic dataset covering nine physical devices spanning five device types (doorbell, baby monitor, security camera, thermostat, webcam). Each device constitutes one federation client; all nine are eligible for calibration. Calibration set sizes range from 2,622 (Ecobee Thermostat) to 35,048 (Philips Baby Monitor) benign samples per device (all ≥ 100 , satisfying the eligibility threshold). Each device’s benign stream is partitioned chronologically into train/calibration/test segments in a 60/1/20/1/18 ratio (the two 1% gaps separate segments to avoid temporal leakage); training and calibration use benign traffic only, and test data, test labels, and training labels are never touched by the attack. The federation trains a per-client autoencoder on benign traffic using FedAvg [14] for feature extraction, then enters the threshold-calibration stage at a fixed quantile $q=0.95$. Table 2 lists the autoencoder architecture, training hyperparameters, and the calibration size range; injection fractions, seeds, and the reservoir are detailed under Attack Parameterization below.

4.2 Threshold Policies

Three threshold-policy families are evaluated, covering the main design points in the personalization literature:

Global. GLOBAL: $\tau_{\text{global}} = \frac{1}{N} \sum_{i=1}^N \tau_i^{\text{local}}$, where $\tau_i^{\text{local}} = \text{quantile}(\mathcal{C}_i, q)$. All clients receive the same threshold; any per-client shift is diluted by $1/N$.

Local. LOCAL: each client retains τ_i^{local} . Shifts are confined to the compromised client; no spillover.

Cluster. CLUSTER: clients are grouped on a standard-scaled 4-feature fingerprint (mean, std, skew, p_{95} of $\mathcal{C}_i$) via k -means++ [1] ($K=3$, `random_state=42`; reassigned each seed). Within each cluster, the shared threshold is the cluster mean; the higher seed-to-seed variance in CLUSTER CV(FPR) reflects cluster-assignment variability. A compromised client may alter the threshold for all co-cluster members (*intra-cluster spillover*); clients in other clusters are unaffected.

4.3 Attack Parameterization

For each (policy, attack objective, injection fraction $f \in \{0.0, 0.1, 0.2, 0.4\}$) combination, we run 10 paired seeds: each seed selects a training configuration and an independent poisoning draw. A clean copy of the calibration array is preserved; only the poisoned copy is used for threshold computation. The four source-objective pairs are: (1) HIGH-SCORE-BENIGN + THRESHOLD-RAISE (primary raising attack), (2) LOW-SCORE-BENIGN + THRESHOLD-LOWER (primary lowering attack), (3) RANDOM-BENIGN + THRESHOLD-RAISE (objective-matched negative control for the raising attack), and (4) RANDOM-BENIGN paired with THRESHOLD-LOWER (negative control for the lowering attack). The attack is applied to each of the nine clients as victim in turn (single-client scope), yielding 4 source-objective pairs $\times$ 3 threshold policies $\times$ 4 injection fractions $\times$ 9 victim clients $\times$ 10 seeds = 4,320 experimental cells total. Training seed $i \in \{0, \dots, 9\}$ is paired with poisoning seed $i+100$ and analysis seed $i+300$, giving statistically independent stochasticity across training, calibration sampling, and bootstrap resampling.

4.4 Statistical Analysis

Gate 1 (material shift, primary gate). Threshold shift $\Delta\tau_v = \tau_v^{\text{poisoned}} - \tau_v^{\text{clean}}$ is computed per seed. A seed is *Gate-1 significant* when $|\Delta\tau_v| > \delta\tau_v$ and the sign is consistent with the attack objective, where the materiality threshold is $\delta\tau_v = \max(0.1 \times \text{IQR}(\mathcal{C}_v^{\text{clean}}), 0.01 \times \text{median}_j(\text{IQR}(\mathcal{C}_j^{\text{clean}})))$. Gate-1 passes when $\geq 8/10$ seeds are significant and the victim-majority condition ($\geq 5/9$ victims significant, i.e. a simple majority of the nine federation clients) holds.

Gate 2 (directional excess over control). Bootstrap 95% CI on mean $\Delta\tau$ excludes zero for the primary strategy but spans zero for the negative control (RANDOM-BENIGN).

Gate 3 (downstream harm). Mean victim TPR change, ΔTPR , is directionally consistent (negative for raising, positive for lowering) across $\geq 8/10$ seeds.

Confidence intervals. All CIs are 95% percentile bootstrap over 10 seed-level aggregates, with $B=10,000$ resamples. Primary claim gates are evaluated over these paired seed-level aggregates; the per-victim-seed cells are descriptive support, not independent units. With ten paired seeds, percentile-bootstrap

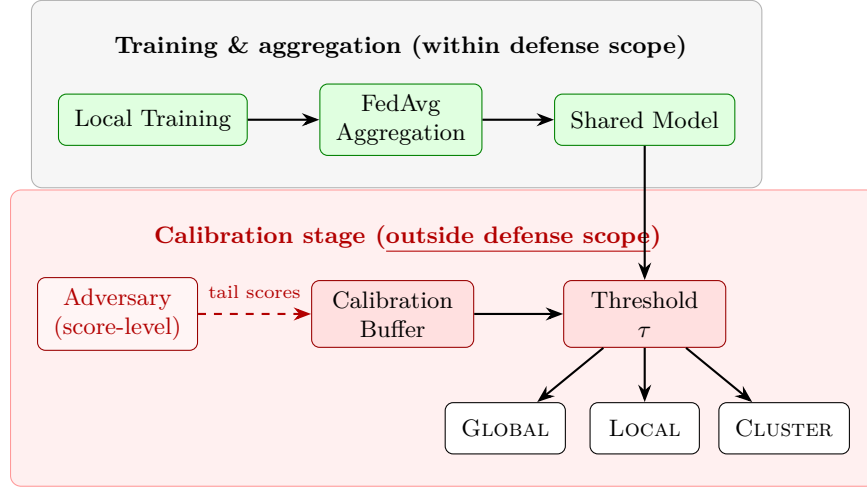


Fig. 1. Attack surface: the threshold-calibration stage occurs after FL training completes. Green boxes mark the training and aggregation phases that existing FL-IDS defenses cover; the red region marks the calibration stage, which those defenses do not observe (no defense is implemented in this study). An adversary with score-level access to one client’s calibration buffer can shift τ without modifying model weights, gradients, or labels.

intervals are approximate uncertainty estimates rather than exact coverage guarantees, and may deviate from nominal at the tails.

Gate constant justification. The 10 % IQR coefficient for Gate-1 materiality ($0.1 \times \text{IQR}$) exceeds the calibration instability observed in the clean baseline while remaining below the expected tail-injection shift at $f=0.1$; the 1 % cross-client floor ($0.01 \times \text{median}_j(\text{IQR})$) prevents degenerate clients with near-zero score spread from counting as material on arbitrarily small shifts. Both constants were locked from clean-artifact statistics before any poisoned run was executed.

4.5 Isolation Sanity Check

To confirm that the attack is confined to the calibration stage, we verify: (i) AU-ROC is invariant under poisoning for all cells, (ii) per-client rank ordering of clean anomaly scores is unchanged, and (iii) model weights before and after calibration are bitwise identical. All three hold in all 4320 cells.

5 Results

5.1 Clean-Baseline Policy Profile

Table 3 reports clean-baseline dispersion statistics across all ten seeds. $\text{CV}(\text{FPR})$, the coefficient of variation of per-client false-positive rates ($\text{ddof}=0$), is lower

when FPR treatment is more uniform across the federation. LOCAL achieves the lowest CV(FPR) (0.333 ± 0.034). GLOBAL exhibits the highest CV(FPR) (0.905 ± 0.039) because a single federation-wide threshold cannot simultaneously fit nine devices with different traffic distributions. CLUSTER lies in between (0.615 ± 0.143) with higher variance across seeds, reflecting sensitivity to cluster assignment. Bootstrap 95% CIs on the pairwise CV(FPR) difference exclude zero: Global–Local = 0.572 [0.537, 0.607]; Global–Cluster = 0.291 [0.214, 0.358]; Cluster–Local = 0.281 [0.201, 0.374], confirming that all three policies are statistically distinguishable on baseline fairness. Figure 2 shows the per-client FPR distributions for GLOBAL and LOCAL under the clean baseline. This fairness gain is not a uniform improvement: LOCAL improves cross-client FPR equity but does not dominate all accuracy-oriented metrics. In particular, GLOBAL retains the best P10 Macro-F1 (0.398 vs. LOCAL 0.335, Table 3), so the worst-tail client is detected better under the federation-mean threshold. We therefore report both fairness-style dispersion (CV(FPR)) and detection-quality (P10 Macro-F1) metrics rather than a single summary.

5.2 Raising Attack: Primary Results

Table 4 and Figure 3(a) report threshold-shift and downstream detection harm for the HIGH-SCORE-BENIGN + THRESHOLD-RAISE strategy.

All three policies pass all three gates. The seed-level Gate-1 pass rate is 100% (10/10 seeds) at every non-zero fraction for all three policies. For LOCAL and CLUSTER, all nine clients are individually significant in every seed. For GLOBAL at $f=0.1$, two devices (Provision PT-737E and PT-838) are intermittently non-significant (2/10 and 4/10 seeds respectively), but the victim-majority condition ($\geq 5/9$ per seed) still holds because the remaining seven clients are consistently significant. Bootstrap CIs on $\Delta\tau$ exclude zero and remain non-overlapping with the RANDOM-BENIGN control, which produces CIs spanning zero, a 0% Gate-1 pass rate, and $|\Delta\text{TPR}| \leq 1$ pp at every fraction and policy (Table 4, Neg. Ctrl rows, reports representative $f=0.4$ values; all fractions are similarly near-null and the complete matrix is archived in the artifact bundle). The targeted effects are therefore attributable to tail contamination, not to position perturbation alone.

Policy-dependent magnitude. The raising attack reveals a clear policy-dependent magnitude profile. LOCAL and CLUSTER exhibit the largest threshold shifts: at $f=0.4$, $\Delta\tau = +0.902$ [+0.864, +0.944] and +0.899 [+0.859, +0.941] respectively. GLOBAL shows a structurally diluted but still significant shift: +0.100 [+0.096, +0.105] at $f=0.4$. The dilution is expected by construction—GLOBAL’s threshold is the federation mean, so a single-client shift is divided by $N=9$ —but the empirical question is whether the residual magnitude ($\approx 10\%$ of the local shift) suffices to produce measurable downstream harm. It does.

Downstream harm (Gate 3). Mean victim ΔTPR is negative and directionally consistent across all seeds for all three policies. At $f=0.4$: LOCAL -7.9 pp, CLUSTER -4.9 pp (plateauing from $f=0.2$), GLOBAL -6.1 pp. The GLOBAL harm (-6.1 pp at the poisoned client) is notable: the shared threshold

means all eight non-victim clients receive the identical $\Delta\tau$ and incur comparable harm (Table 5), unlike LOCAL/CLUSTER where non-victim harm is bounded.

5.3 Lowering Attack: Secondary Results (Conditional)

Figure 3(b) shows threshold shift for the LOW-SCORE-BENIGN + THRESHOLD-LOWER strategy. The adversary’s goal is to increase the false alarm rate; the $\Delta\text{TPR} > 0$ below is a threshold-displacement indicator, not a security benefit. Elevated alarm burden may lead operators to disable the anomaly detector.

Local and Cluster succeed; Global is marginal. LOCAL achieves $\Delta\tau = -0.164$ $[-0.176, -0.154]$ at $f=0.4$ with elevated FPR disparity ($\Delta\text{CV}(\text{FPR}) = +0.053$); Gate-3 $\Delta\text{TPR} = +11.8$ pp confirms threshold displacement. CLUSTER similarly achieves -0.154 $[-0.167, -0.142]$ at $f=0.4$ with $\Delta\text{CV}(\text{FPR}) = +0.060$ and $\Delta\text{TPR} = +10.3$ pp. GLOBAL is structurally bounded: at $f=0.4$, $\Delta\tau = -0.018$ $[-0.020, -0.017]$, Gate-1 passes only at $f \geq 0.2$, and $\Delta\text{TPR} = +2.1$ pp.

FPR disparity (operational harm). The operational harm metric is $\Delta\text{CV}(\text{FPR})$ (change in coefficient of variation of per-client FPR, dimensionless): $+0.053$ (LOCAL, 10/10 seeds) and $+0.060$ (CLUSTER, 8/10 seeds) at $f=0.4$, while GLOBAL shows -0.019 (shared-threshold proportional FPR rise reduces relative disparity).

Mechanism and conditionality. Lowering a $q=0.95$ quantile requires displacing scores in the upper-tail support; random position replacement achieves this with probability $\approx f$ (40 % at $f=0.4$), leaving 60 % of upper-tail support intact. The GLOBAL policy compounds this via $1/N$ dilution. We present these results as a lower bound under REPLACE-FIXED-BUDGET; targeted upper-tail removal would achieve larger shifts.

5.4 Policy-Differentiation Summary

Table 5 quantifies the blast-radius contrast under the raising attack at $f=0.4$. GLOBAL propagates the threshold shift to all nine clients: the non-victim $\Delta\tau$ equals the victim $\Delta\tau$ by construction (shared threshold), and blast encompasses $\approx 8/8$ non-victims in nearly every seed. CLUSTER achieves near-LOCAL victim-level harm; non-victim threshold changes are operationally negligible (mean $\Delta\tau \approx 0.000$, driven by K-means cluster reassignment rather than aggregation), with 6/8 non-victims showing statistically significant but small-magnitude shifts. The blast count measures threshold significance, not downstream harm: these significant-but-negligible CLUSTER shifts do not produce measurable non-victim detection-rate change, unlike GLOBAL, where the shared threshold transmits the victim’s $\Delta\tau$ directly to every non-victim. LOCAL yields the highest per-victim shift with zero non-victim blast by policy design.

A policy designer must choose between isolated per-victim exposure (LOCAL) and diluted but fleet-wide propagation (GLOBAL), with CLUSTER carrying churn-level effects without direct aggregation spillover.

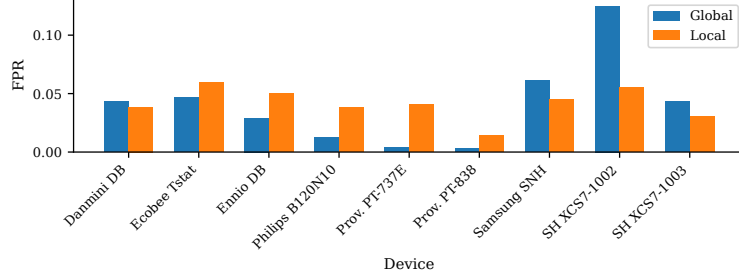


Fig. 2. Per-client FPR under clean baseline, GLOBAL vs. LOCAL threshold (seed 0; CV(FPR) values across all 10 seeds reported in Table 3). GLOBAL forces a single threshold on nine devices with heterogeneous score distributions, producing CV(FPR) = 0.87 for seed 0 (mean = 0.905 ± 0.039 across seeds). LOCAL fits each client’s own calibration distribution, achieving CV(FPR) = 0.33 on average. CLUSTER is omitted here; its baseline is in Table 3 (CV(FPR) = 0.615 ± 0.143).

6 Discussion and Limitations

Score-level proxy and detectability. The experiment operates at the score level: reconstruction scores in the calibration buffer are replaced programmatically rather than through generated traffic. A complete end-to-end attack would additionally require generating network packets whose autoencoder reconstruction error lands in the target tail of the victim’s score distribution—a non-trivial requirement spanning feature engineering, the encoder-decoder nonlinearity, and physical network behavior. We present this work as a *vulnerability-surface characterization*, not a deployed attack; traffic-level realizability is future work. A related implementation limitation: with-replacement sampling from the tail reservoir at $f=0.4$ produces duplicate scores in the calibration buffer. With reservoir size $R \approx 10\% \cdot n_v$ and the smallest per-client calibration set ($n_v \approx 2,622$, distinct from the federation size $N=9$), the expected collision count $(n_v \cdot f)^2 / (2R) \approx 2,100$ expected collision pairs at $f=0.4$ —easily flagged by a uniqueness check; evading requires score perturbation at the cost of leaving the exact tail boundary. We flag detectability as an open problem.

GLOBAL spillover, lowering bounds, and policy implications. The $1/N$ -diluted GLOBAL shift is definitional; what is empirical is that the residual magnitude ($\approx 10\%$ of the local shift, $\Delta\text{TPR} = -6.1$ pp at $f=0.4$) remains measurable above the seed-to-seed variance. The Local/Cluster lowering results are a *lower bound* under REPLACE-FIXED-BUDGET (§5): adversaries with access to the buffer’s rank structure could achieve larger threshold displacements and alarm-burden effects. These results reveal a structural trade-off in policy design: LOCAL maximizes per-client isolation but maximizes per-victim exposure, whereas GLOBAL limits per-victim shift via averaging but propagates residual harm federation-wide; defense designers must choose which risk to bound.

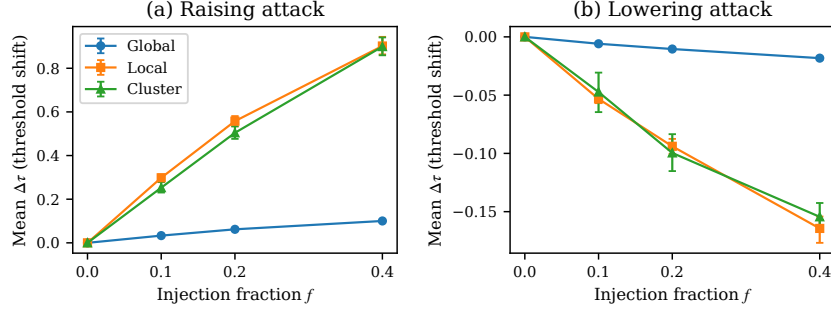


Fig. 3. Mean threshold shift $\Delta\tau$ vs. injection fraction f for the raising (a) and lowering (b) attacks across three policies. Error bars: 95 % bootstrap CI over 10 seeds. GLOBAL exhibits the smallest magnitude in both directions due to $1/N$ federation averaging. RANDOM-BENIGN negative control (not shown) produces CIs spanning zero and $G1=0\%$ at all fractions and policies; representative values at $f=0.4$ are in Table 4 (Neg. Ctrl rows).

Table 3. Clean-baseline dispersion statistics per threshold policy (N-BaIoT, $n=10$ seeds, 9 clients). CV: coefficient of variation (ddof=0; lower is more uniform). Worst BA: minimum per-client balanced accuracy (higher better). P10 Macro-F1: 10th-percentile client macro-F1 (higher better). Bold = best per column. * Actual std ≈ 0.0002 ; rounds to zero at three decimals.

Policy	CV(FPR)	CV(TPR)	Worst BA	P10 Macro-F1
GLOBAL	0.905 ± 0.039	0.255 ± 0.030	0.657 ± 0.008	0.398 ± 0.053
LOCAL	0.333 ± 0.034	0.296 ± 0.036	0.655 ± 0.005	0.335 ± 0.058
CLUSTER	0.615 ± 0.143	0.313 ± 0.009	0.656 ± 0.004	$0.299 \pm 0.000^*$

Scope limitations. Claims are bounded to N-BaIoT (9 clients; generalizability to other federated IoT datasets and larger federation sizes is open), autoencoder architecture, FedAvg, and $q=0.95$ (quantile sensitivity is future work). The experiment assumes a single compromised client; multi-client collusion is a natural extension. Generalizability to other anomaly detectors, FL algorithms, and heterogeneous federation profiles is not established. The $n=10$ seed bootstrap CI coverage may differ from nominal at the tails. Defense design for the calibration channel (threshold validation, calibration-set integrity constraints) is out of scope and merits independent study.

Table 4. Threshold shift and detection harm under calibration-stage poisoning (N-BaIoT, $n=10$ seeds). $\Delta\tau$: mean threshold shift [95 % bootstrap CI]. G1: fraction of 10 seeds passing Gate-1. $\Delta\text{TPR (pp)}$: victim detection-rate change; negative for Raise (harm), positive for Lower (alarm-burden increase, not a security benefit). Lowering rows ($\dagger$): $\Delta\text{CV(FPR)}$ is the primary harm metric; ΔTPR confirms threshold displacement. NEG. CONTROL: RANDOM-BENIGN at $f=0.4$; all fractions are similarly near-null; bold=zero G1.

Policy	Attack	f	$\Delta\tau$ [95 % CI]	G1	$\Delta\text{TPR (pp)}$
GLOBAL	Raise	0.1	+0.033 [+0.031, +0.035]	100 %	−2.4
		0.2	+0.062 [+0.059, +0.065]	100 %	−4.6
		0.4	+0.100 [+0.096, +0.105]	100 %	−6.1
LOCAL	Raise	0.1	+0.298 [+0.281, +0.314]	100 %	−6.0
		0.2	+0.558 [+0.535, +0.581]	100 %	−7.3
		0.4	+0.902 [+0.864, +0.944]	100 %	−7.9
CLUSTER	Raise	0.1	+0.253 [+0.231, +0.277]	100 %	−3.9
		0.2	+0.504 [+0.475, +0.532]	100 %	−4.9
		0.4	+0.899 [+0.859, +0.941]	100 %	−4.9
GLOBAL	Lower $\dagger$	0.1	−0.006 [−0.006, −0.006]	60 %	+0.8
		0.2	−0.010 [−0.011, −0.010]	100 %	+1.1
		0.4	−0.018 [−0.020, −0.017]	100 %	+2.1
LOCAL	Lower $\dagger$	0.1	−0.053 [−0.057, −0.049]	100 %	+2.8
		0.2	−0.094 [−0.100, −0.088]	100 %	+5.9
		0.4	−0.164 [−0.176, −0.154]	100 %	+11.8
CLUSTER	Lower $\dagger$	0.1	−0.047 [−0.065, −0.031]	100 %	+2.2
		0.2	−0.100 [−0.115, −0.084]	100 %	+5.4
		0.4	−0.154 [−0.167, −0.142]	100 %	+10.3
GLOBAL	Neg. Ctrl	0.4	−0.000 [−0.001, +0.001]	0 %	+0.3
LOCAL	Neg. Ctrl	0.4	−0.002 [−0.009, +0.006]	0 %	+0.9
CLUSTER	Neg. Ctrl	0.4	−0.009 [−0.022, +0.003]	0 %	+0.4

Table 5. Spillover profile under the raising attack ($f=0.4$, $n=10$ seeds). Blast: mean fraction of non-victim clients with a statistically significant threshold shift. $\dagger$ Non-victim threshold equals victim threshold by construction (shared global threshold); expected non-victim $\Delta\text{TPR} \approx$ victim ΔTPR . $\ddagger$ Effect driven by K-means cluster reassignment (churn); mean non-victim $\Delta\tau=0.0003$, std = 0.017.

Policy	Victim $\Delta\tau$	Victim $\Delta\text{TPR (pp)}$	Non-vic. $\Delta\tau$	Blast
GLOBAL	+0.100	−6.1	+0.100 $\dagger$	$\approx 8/8$
CLUSTER	+0.899	−4.9	$\approx 0.000\ddagger$	$\approx 6/8$
LOCAL	+0.902	−7.9	0.000	0/8

7 Conclusion

We characterized the post-training threshold-calibration stage in federated IoT anomaly detection as an attack surface that training-phase and aggregation-phase defenses do not address. Under a score-level contamination model on the N-BaIoT nine-device federation (FedAvg autoencoder, $q=0.95$, single compromised client), the raising attack degrades victim detection rates by 2.4–7.9 pp across non-zero injection fractions (4.9–7.9 pp at $f=0.4$) with 100% Gate-1 pass rates and 95% bootstrap CIs excluding zero; these intervals are descriptive uncertainty estimates over ten paired seeds. The lowering attack shifts thresholds downward, increasing victim TPR by 10–12 pp and $\Delta\text{CV}(\text{FPR})$ by 0.05–0.06 (lower bound under REPLACE-FIXED-BUDGET) for LOCAL and CLUSTER. The policy ordering—per-victim exposure maximized under LOCAL, diluted but federation-wide under GLOBAL, intra-cluster under CLUSTER—implies calibration-stage hardening must match the aggregation design, not only the training-phase defense. These findings isolate a necessary calibration-channel vulnerability mechanism, not a complete traffic-generation attack. Future work should address traffic-level realizability, calibration-stage integrity defenses, and generalization across detector architectures and datasets.

Reproducibility. All experiments are pipeline-generated and locally archived (code commit [f9faf94](#)). The artifact bundle comprises the experiment configuration; the seed scheme (training 0–9, paired poisoning 100–109, analysis/bootstrap 300–309); the split manifest with per-device calibration sizes and the chronological benign-only 60/1/20/1/18 split semantics; the raw per-cell results; negative-control records; isolation sanity checks; and figure/table scripts. Code and artifacts will be released publicly on acceptance and are available to reviewers on request.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Arthur, D., Vassilvitskii, S.: k-means++: The advantages of careful seeding. In: Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA). pp. 1027–1035 (2007)
2. Bagdasaryan, E., Veit, A., Hua, Y., Estrin, D., Shmatikov, V.: How to backdoor federated learning. In: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS). pp. 2938–2948. PMLR (2020)
3. Barreno, M., Nelson, B., Joseph, A.D., Tygar, J.D.: The security of machine learning. *Machine Learning* **81**(2), 121–148 (2010)
4. Bhagoji, A.N., Chakraborty, S., Mittal, P., Calo, S.: Analyzing federated learning through an adversarial lens. In: Proceedings of the 36th International Conference on Machine Learning (ICML). pp. 634–643. PMLR (2019)
5. Biggio, B., Nelson, B., Laskov, P.: Poisoning attacks against support vector machines. In: Proceedings of the 29th International Conference on Machine Learning (ICML). pp. 1467–1474 (2012)

6. Blanchard, P., El Mhamdi, E.M., Guerraoui, R., Stainer, J.: Machine learning with adversaries: Byzantine tolerant gradient descent. In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 30 (2017)
7. Cao, X., Fang, M., Liu, J., Gong, N.Z.: FLTrust: Byzantine-robust federated learning via trust bootstrapping. In: *Network and Distributed System Security Symposium (NDSS)* (2020)
8. Cretu, G.F., Stavrou, A., Locasto, M.E., Stolfo, S.J., Keromytis, A.D.: Casting out demons: Sanitizing training data for anomaly sensors. In: *2008 IEEE Symposium on Security and Privacy*. pp. 81–95. IEEE (2008)
9. Geiping, J., Bauermeister, H., Dröge, H., Moeller, M.: Inverting gradients—how easy is it to break privacy in federated learning? In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 33, pp. 16937–16947 (2020)
10. Geyer, R.C., Klein, T., Nabi, M.: Differentially private federated learning: A client level perspective. In: *NeurIPS Workshop on Private Multi-Party Machine Learning* (2017)
11. Ghosh, A., Chung, J., Yin, D., Ramchandran, K.: An efficient framework for clustered federated learning. In: *Advances in Neural Information Processing Systems (NeurIPS)*. vol. 33, pp. 19586–19597 (2020)
12. Kloft, M., Laskov, P.: Online anomaly detection under adversarial impact. In: *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics (AISTATS)*. pp. 405–413. PMLR (2012)
13. Li, T., Sahu, A.K., Talwalkar, A., Smith, V.: Federated learning: Challenges, methods, and future directions. *IEEE Signal Processing Magazine* **37**(3), 50–60 (2020)
14. McMahan, B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*. pp. 1273–1282. PMLR (2017)
15. Meidan, Y., Bohadana, M., Mathov, Y., Mirsky, Y., Shabtai, A., Breitenbacher, D., Elovici, Y.: N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Computing* **17**(3), 12–22 (2018)
16. Nguyen, D.C., Pham, Q.V., Pathirana, P.N., Ding, M., Seneviratne, A., Lin, Z., Dobre, O., Hwang, W.J.: Federated learning for smart healthcare: A survey. *ACM Computing Surveys* **55**(3), 1–37 (2022)
17. Rey, V., Sanchez Sanchez, P.M., Huertas Celdrán, A., Bovet, G.: Federated learning for malware detection in IoT devices. *Computer Networks* **204**, 108693 (2022)
18. Sattler, F., Müller, K.R., Samek, W.: Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE Transactions on Neural Networks and Learning Systems* **32**(8), 3710–3722 (2020)
19. Tolpegin, V., Truex, S., Gursoy, M.E., Liu, L.: Data poisoning attacks against federated learning systems. In: *25th European Symposium on Research in Computer Security (ESORICS)*. pp. 480–501. Springer (2020)
20. Yin, D., Chen, Y., Kannan, R., Bartlett, P.: Byzantine-robust distributed learning: Towards optimal statistical rates. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. pp. 5650–5659. PMLR (2018)
21. Zhou, C., Paffenroth, R.C.: Anomaly detection with robust deep autoencoders. In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 665–674 (2017)